

Small-step Semantics of *SimpleExp*

$$\rightarrow \subseteq \text{SimpleExpr} \times \text{SimpleExpr}$$

$$(S\text{-LEFT}) \quad \frac{E_1 \rightarrow E'_1}{E_1 +_s E_2 \rightarrow E'_1 +_s E_2}$$

$$(S\text{-RIGHT}) \quad \frac{E \rightarrow E'}{n +_s E \rightarrow n +_s E'}$$

$$(S\text{-ADD}) \quad \frac{}{n_1 +_s n_2 \rightarrow n_3} \quad n_3 = n_1 + n_2$$